



CephalFormer: Incorporating Global Structure Constraint into Visual Features for General Cephalometric Landmark Detection

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Abstract. Accurate cephalometric landmark detection is a crucial step in orthodontic diagnosis and therapy planning. However, existing deep learning-based methods lack the ability to explicitly model the complex dependencies among visual features and landmarks. Therefore, they fail to adaptively encode the landmark’s global structure constraint into the representation of visual concepts and suffer from large biases in landmark localization. In this work, we propose CephalFormer, which exploits the correlations between visual concepts and landmarks to provide meaningful guidance for accurate 2D and 3D cephalometric landmark detection. CephalFormer explores local-global anatomical contents in a coarse-to-fine fashion and consists of two stages: (1) a new efficient Transformer-based architecture for coarse landmark localization; (2) a novel paradigm based on self-attention to represent visual clues and landmarks in one coherent feature space for fine-scale landmark detection. We evaluated CephalFormer on two public cephalometric landmark detection benchmarks and a real-patient dataset consisting of 150 skull CBCT volumes. Experiments show that CephalFormer significantly outperforms the state-of-the-art methods, demonstrating its generalization capability and stability to naturally handle both 2D and 3D scenarios under a unified framework.

Keywords: Cephalometric landmark detection · Transformer

1 Introduction

Accurate anatomical landmark detection is a fundamental procedure in cephalometric analysis, which is widely adopted in orthodontic diagnosis, treatment

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planning and evaluation [7]. In clinical practice, manual annotation is typically tedious and subjective [19]. Therefore, automatic and accurate landmark detection has attracted great interest from both academia and industry.

Conventionally, cephalometric analysis is performed on 2D cephalograms. In recent years, convolutional neural networks (CNNs) have achieved significant success in 2D cephalometric landmark detection [1, 12–14, 25, 27, 29]. Most existing methods design complex network architectures or use powerful mechanisms such as self-attention to improve the detection accuracy. For example, Chen *et al.* [3] proposed an attentive feature pyramid fusion module (AFPF) to shape semantically enhanced features for landmark localization. Zhou *et al.* [27] developed a learning-to-learn framework to search objective metrics for regressing multiple landmark heatmaps dynamically. Zeng *et al.* [25] proposed a cascaded three-stage CNNs for cephalometric landmark detection.

Variations on individual structures, appearance ambiguity, and image complexity are the main factors that hinder accurate landmark detection. A potential solution, as described in [12, 13], is to incorporate prior knowledge on spatial structure configurations of landmarks into the learned heatmaps, which could enable the network to simplify the modeling of the underlying distribution of anatomical features. More recently, Li *et al.* [10] proposed a deep adaptive graph learning approach to model landmarks as a graph to exploit prior structural knowledge. Despite popularity and success, these methods only explore the relationships among landmarks via additional modules or GCNs [8] and fail to represent input image features and landmarks in one coherent graph. As a result, they can not explicitly model the dependencies between all visual features and anatomical landmarks. This major drawback makes them susceptible to large anatomical variations and imperfect overlaps of anatomical structures. In practice, The landmarks are usually associated with anatomical features [22]. Building the correlations between visual concepts and landmarks can help the network adaptively learn the global structure constraint on spatial configurations of landmarks. So far, learning explicit dependencies between all features and anatomical landmarks via representing them jointly in one coherent feature space has not yet been investigated.

Another challenge in this field is the lack of an efficient and accurate 3D cephalometric landmark detection methodology. 2D cephalograms only provide a projected 2D view of real 3D anatomical structures, resulting in problems such as geometric distortions and superimpositions [24]. However, the expensive computational cost and complex search space make it impractical to transfer the state-of-the-art methods from 2D to 3D directly.

Several research efforts have been conducted to explore 3D cephalometric landmark detection. Lang *et al.* [9] proposed a two-stage coarse-to-fine method, utilizing 3D mask R-CNN [5] and GCNs to detect landmarks. However, it takes minutes to infer a CBCT volume and lacks the consideration of the global structure constraint. Chen *et al.* [4] also proposed a two-stage coarse-to-fine framework based on LSTM [6] for 3D landmark detection. However, it still relies on an additional graph attention module to encode the landmark's global struc-

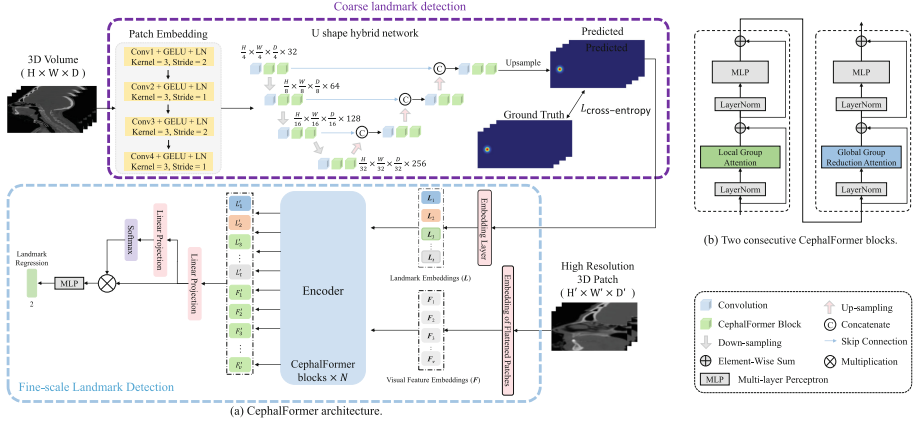


Fig. 1. Overview of CephalFormer. It first locates the coarse landmarks via heatmap regression and then performs fine-scale landmark detection. These two stages are shown in the purple and blue dashed box respectively. (Color figure online)

ture constraint and fails to input the image features and landmark embeddings jointly. Moreover, the recursive prediction process of LSTM is slow and suffers from the vanishing gradient problem.

Motivated by the aforementioned limitations, we propose a general end-to-end deep learning method, CephalFormer, for automatic 2D and 3D cephalometric landmark detection. The contribution of our work is threefold. (1) We propose the first general Transformer-based framework that naturally handles both 2D and 3D scenarios. (2) We are the first to study how to represent visual features and landmarks into a coherent feature space to explicitly incorporate the global structure constraint for accurate cephalometric landmark detection. (3) Our method significantly outperforms the state-of-the-art methods on two public cephalometric landmark detection benchmarks and a real-patient dataset.

2 Methods

As shown in Fig. 1(a), CephalFormer adopts a coarse-to-fine two-stage strategy. In the first stage, a new U shape hybrid structure is built based on the combination of interleaved convolution and CephalFormer blocks (see Fig. 1(b)) for coarse landmark localization. In the second stage, we input the visual features of high-resolution cropped patches and landmark embeddings jointly into a Transformer encoder, which allows rich information exchange and adaptively incorporates the global structural constraint for accurate coordinate refinement. The two-stage models are trained end-to-end. In the following, we take the 3D scene as an example to present details of these two stages and give more insights into our method. 2D scenes can be easily handled by replacing configurations such as 3D convolution operations with 2D ones.

we propose GGRA to enable cross-group information exchange. As shown in Fig. 2(b), we first downsample each group to a single representative to summarize the important information by a spatial reduction module, which consists of two convolutional layers and a global average pooling layer. Then we feed the representative of each group to the multi-head self-attention layer to compute queries, keys and values. To further reduce computation cost, we use subsampled keys and values to communicate with other groups. Finally, the patch tokens are upsampled by a spatial expansion module, which consists of two deconvolutional layers. GGRA reduces the cost of self-attention to $O(\frac{m_i^2 n_i^2 d_i^2}{\alpha^3} C_i)$. The coarse landmark detection is accomplished via heatmap regression on the output feature maps of the U shape hybrid network.

2.2 Fine-Scale Coordinate Refinement

Following most two-stage coarse-to-fine methods [4, 9], we sample high resolution 3D patches $P \in \mathbb{R}^{H' \times W' \times D' \times C}$ in the vicinity of the landmark locations estimated in the first stage. There is no mechanism to encode interactions between visual information and prior knowledge on landmark topology in one joint feature space in traditional models. To address this drawback, we propose a technique to easily incorporate landmark configurations into visual features.

First, we split P into $m' \times n' \times d'$ fixed-size patches and linearly embed each of them to obtain a set of visual feature embeddings $F = \{F_1, F_2, \dots, F_v\}$, $F_j \in \mathbb{R}^g$, $v = m' \times n' \times d'$. Then we consider each vector $L_k \in \mathbb{R}^g$ from L , with k ranging from 1 to t (where t is the number of landmarks), to be representative of the k th possible landmark. Landmark embeddings are learned from a linear embedding layer which takes features extracted from the coarse predicted heatmaps as inputs to encode the information of each landmark. We concatenate F and L , add position embeddings and feed the resulting sequence of vectors to a Transformer encoder built with N CephalFormer blocks. Compared with previous methods [10] that only model landmarks as a graph, our Transformer encoder can be viewed as a fully connected graph, which is able to learn any relationships between visual features and landmarks. In this way, landmarks are associated with anatomical features explicitly and the network can adaptively learn the global structure constraint on the spatial configuration of landmarks. Such an advantage is very important for cephalometric landmark detection, because the feasible location of a landmark is not distributed uniformly in image space, but is constrained by the locations of other anatomical landmarks [13].

At last, an attention module, which consists of two linear layers and a softmax layer, is utilized to extract the weight matrix for the output token embeddings (F' and L'). This attention module helps to fully utilize the contributions of all output tokens while distinguishing their differences. Finally, we assign the attention weights to the learned token embeddings and feed them to a multi-layer perceptron to accomplish the final landmark regression. If the model in the first stage misses some landmarks in the prediction, the second-stage model still can get them back since the second-stage model has direct access to the input high resolution image patches.

3 Experiments

3.1 Settings

Dataset: In the 2D scenario, we evaluate our method on the public benchmark from the IEEE ISBI 2015 cephalometric landmark detection Challenge [18, 19]. It contains 400 cephalograms with resolution of 1935×2400 . 150 images are used as training set, the rest 150 images and 100 images are used as test set 1 and test set 2. Each cephalometric image contains 19 landmarks. The size of each pixel is 0.1 by 0.1 mm. We follow [3] to resize the Cephalometric X-rays to 800×640 .

For 3D tasks, we make experiments on two datasets. One is the Public Domain Database for Computational Anatomy (PDDCA) dataset from MIC-CAI Head-Neck Challenge 2015 [15]. It contains 48 patient CT images, of which 33 CT images are labeled with 5 bony landmarks. The dataset is equally distributed into three folds to perform three-fold cross-validation, and the average accuracy of three folds is reported. Following [4], we standardize the voxel spacing to 1 mm. The average resolution is $518 \times 518 \times 384$ after standardization. Besides, an in-house skull dataset is also used for evaluation, which consists of 150 skull CBCT in DICOM format with resolution of $512 \times 512 \times 512$. The isotropic voxel spacing is 0.3 mm. Each data has 33 manually annotated landmarks on the skull. We split the dataset into three folds with 50, 50 and 50 samples to perform three-fold cross-validation. We report the average accuracy of three folds.

Metrics: We follow the original evaluation protocol designated by the official challenge. The evaluation metrics are the mean radial error (MRE), standard deviation (SD), and the successful detection rate (SDR) under 2 mm, 2.5 mm, 3 mm and 4 mm [19]. Besides, we also report the inference time, model parameters and floating point operations (FLOPs) for efficiency comparison.

Implementation Details: We build two architecture variants for 2D and 3D tasks on Pytorch, respectively. For 2D tasks, the input image size in the first and second stage are both 800×640 . In the encoder of the U shape hybrid architecture, the group size of LGA is gradually decreased from 32×24 , 16×12 , 8×6 to 4×3 . The encoder and the decoder are symmetric. In the fine-scale detection stage, the number of CephalFormer blocks N is set as 12 while the embedding dimension g is set as 96. For 3D tasks, the input volume size in the first and second stage are both $518 \times 518 \times 384$ for PDDCA dataset ($512 \times 512 \times 512$ for in-house skull dataset). The group size of LGA is gradually decreased from $32 \times 32 \times 24$, $16 \times 16 \times 12$, $8 \times 8 \times 6$ to $4 \times 4 \times 3$. In the fine-scale detection stage, N is set as 8 while g is set as 64. We adopt cross-entropy loss for heatmap regression and L_1 loss for coordinate regression. The total loss is a weighted sum of these two losses. All experiments are performed on 8 V100 GPUs. We employ an AdamW optimizer for 1000 epochs using a cosine decay learning rate scheduler and 20 epochs of linear warm-up. A mini-batch size of 8, an initial learning rate of 0.001, and a weight decay of 0.05 are used. We use gradient clipping with a max norm of 1.0 to stabilize the training

Table 1. Evaluation metrics of different models on ISBI dataset. * represents the performances copied from the original paper. - represents that no experimental results can be found in the original paper. The best results are in bold.

Model	Time (S)	FLOPs (G)	#param. (M)	Test 1 dataset				Test 2 dataset					
				MRE (SD)	2 mm	2.5 mm	3 mm	4 mm	MRE (SD)	2 mm	2.5 mm	3 mm	4 mm
Arik <i>et al.</i> [1]*	–	–	–	–(-)	75.37	80.91	84.32	88.25	–(-)	67.68	74.16	79.11	84.63
Qian <i>et al.</i> [14]*	1.04	10.5	42.5	1.28(1.17)	82.51	86.25	89.30	90.62	1.54(1.05)	72.40	76.15	79.65	85.90
Chen <i>et al.</i> [3]*	0.07	–	–	1.17(-)	86.67	92.67	95.54	98.53	1.48(-)	75.05	82.84	88.53	95.05
DRM [26]	–	–	–	1.12(0.88)	86.91	91.82	94.88	97.90	1.42(0.84)	76.00	82.90	88.74	94.32
Cascaded [25]*	2.03	0.6	69.1	1.34(0.92)	81.37	89.09	93.79	97.86	1.64(0.91)	70.58	79.53	86.05	93.32
Zhou <i>et al.</i> [27]*	–	–	–	–(-)	–	–	–	–	1.39(1.32)	76.11	84.21	88.79	94.84
Swin-CE [23]*	1.74	56.7	88.3	1.13(-)	87.31	92.93	95.87	98.74	1.44(-)	76.23	83.36	89.15	95.24
GU2Net [29]	0.22	0.5	4.4	1.12(1.27)	87.21	92.55	95.60	97.70	1.37(1.16)	74.11	85.05	89.39	94.04
SCN [13]	0.29	9.2	29.6	1.34(1.22)	81.47	89.36	93.15	97.01	1.65(1.12)	69.94	78.84	85.74	93.89
DACFL [12]*	0.15	–	–	1.18(1.01)	86.20	91.20	94.41	97.68	1.47(0.82)	75.92	83.44	89.27	94.73
DAG [10]	0.67	0.8	23.9	1.04(0.95)	88.49	93.12	95.72	98.42	1.43(0.80)	76.57	83.68	88.21	94.31
CephalFormer-2D	0.10	2.9	25.1	0.92(0.84)	89.79	94.11	96.33	98.94	1.26(0.75)	80.44	87.63	91.40	96.29

process. Augmentations such as rotation, scaling, gaussian blur, and mirroring are utilized during the training process.

3.2 Comparisons with State-of-the-Art Methods

The results on ISBI dataset are shown in Table 1. Our method significantly outperforms the state-of-the-art methods. Especially in the precision of 2 mm, the improvements of SDR on two ISBI test datasets are 1.3% and 3.9%, respectively. Moreover, our CephalFormer-2D behaves much better than SCN, DACFL and DAG. These three methods also incorporate prior global structural constraint but use additional modules or GCNs. CephalFormer differs from these approaches in how it models global structural constraint. Above all, previous methods using graphs or additional CNN modules to incorporate structural prior knowledge do not allow for interactions between visual clues and landmarks while CephalFormer is capable of learning any relationship between visual features and landmarks. Compared with Swin-CE [23], which combines Swin Transformer [11] with a CNN encoder for 2D cephalometric landmark detection, our CephalFormer-2D outperforms it with 3x fewer parameters and 18x lower FLOPs. Swin Transformer also models dependencies within local sub-windows, but it cannot model interactions between different windows. While in our GGRA, sub-groups are connected in an efficient way, which significantly improves the detection performance.

The results on PDDCA dataset and in-house skull dataset are shown in Table 2. PDDCA dataset has much fewer samples (33 samples vs. 150 samples) than our in-house dataset. However, our CephalFormer-3D is robust to such a small dataset and significantly outperforms other methods. The SDRs are increased by 14% in the precision of 2 mm. As for our in-house dataset, the average error of three-fold cross-validation is 1.33 mm, and 82.11% of landmarks are within the 2 mm clinically acceptable errors. The inference time of our method on these two datasets is 0.14 and 0.33 s, respectively. CephalFormer-3D is significantly faster and more efficient than the other two-stage methods and has much

Table 2. Comparison with state-of-the-art methods on 3D tasks.

Model	#param. (M)	PDDCA dataset							In-house skull dataset						
		Time (S)	FLOPs (G)	MRE (SD)	2 mm	2.5 mm	3 mm	4 mm	Time (S)	FLOPs (G)	MRE	2 mm	2.5 mm	3 mm	4 mm
LA-GCN [9]	288.1	93.21	107.4	3.23 (2.52)	35.68	46.76	58.19	69.48	287.38	222.9	2.13(1.72)	67.22	70.01	79.33	85.17
SA-LSTM [4]	75.9	0.27	26.5	2.37 (1.60)	56.36	71.60	80.00	89.99	1.29	68.1	1.81(1.27)	72.57	80.46	84.69	88.62
CephalFormer-3D	35.3	0.14	8.3	1.85(1.44)	70.31	80.29	88.16	92.34	0.33	26.4	1.33 (1.07)	82.11	89.53	93.45	96.77

Table 3. Ablation study of network components.

Model	ISBI test 2 dataset					PDDCA dataset				
	MRE (SD)	2 mm	2.5 mm	3 mm	4 mm	MRE (SD)	2 mm	2.5 mm	3 mm	4 mm
U-Net [16]	1.62 (1.46)	70.11	75.61	80.20	85.39	3.86 (2.55)	30.14	39.26	47.92	60.88
nnFormer [28]	1.63 (1.55)	69.13	79.66	86.19	90.51	3.39 (2.64)	32.91	44.73	56.98	67.53
ϕ_c	1.45 (1.38)	76.29	84.33	88.80	94.82	2.33 (1.68)	55.48	70.87	79.16	89.10
$\phi_c + GCNs$	1.39 (0.77)	77.81	85.05	89.10	95.14	2.11 (1.57)	62.29	74.33	83.75	90.08
CephalFormer	1.26 (0.75)	80.44	87.63	91.40	96.29	1.85 (1.44)	70.31	80.29	88.16	92.34

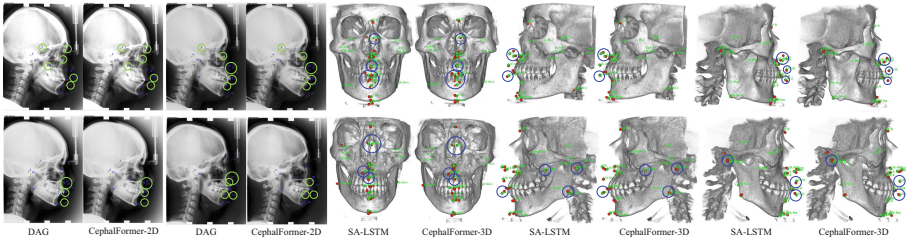


Fig. 3. Qualitative comparison of different models on ISBI dataset and our in-house skull dataset. The red points are the learned landmarks while the green points are the ground truth labels. We mark significant improvements with circles. It is clear that our model’s results have more overlap regions of red and green points than other methods. (Color figure online)

fewer parameters and lower FLOPs. The qualitative comparison of CephalFormer and suboptimal methods is shown in Fig. 3. Note that directly transferring 2D methods to 3D leads to deficient performance, so their results are not reported.

3.3 Ablation Study

We perform ablation study on the two public benchmarks. First, in order to demonstrate the effectiveness of our LGA and GGRA, we replace the U shape hybrid structure in the first stage with several other close alternatives, *i.e.*, U-Net [16] and nnFormer [28]. Second, we evaluate the performance on ϕ_c (with coarse landmark detection network only), $\phi_c + GCNs$ (with GCNs to incorporate the global structural constraint for fine-scale coordinate refinement), and CephalFormer to demonstrate the effectiveness of modeling interactions between visual features and landmarks via the Transformer encoder in the second stage.

As shown in Table 3, when comparing CephalFormer with U-Net and nnFormer, it is evident that our U shape hybrid structure consisting of LGA and GGRA improves the performance by a large margin. This is attributed to the efficiency and effectiveness of our new self-attention mechanism, which enables whole-volume inference, whereas U-Net lacks the ability to model long-range dependencies and patch-based nnFormer has limited receptive fields. By comparing CephalFormer with ϕ_c and $\phi_c + GCNs$, we observe much improvement of detection accuracy, which demonstrates the effectiveness of embeddings learned with the dependencies between visual features and landmarks.

4 Conclusion

In this paper, we propose a novel end-to-end Transformer-based framework called CephalFormer for 2D and 3D cephalometric landmark detection. It first locates the coarse landmarks and then refines them via exploring complex dependencies between visual features and landmarks. Our method adaptively learns the global structure constraint on the spatial configuration of landmarks and thus significantly improves the cephalometric landmark detection accuracy in the clinically accepted precision range of 2.0 mm.

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References

1. Arik, S.Ö., Ibragimov, B., Xing, L.: Fully automated quantitative cephalometry using convolutional neural networks. *J. Med. Imaging* **4**(1), 014501 (2017)
2. Chen, J., et al.: TransUNet: transformers make strong encoders for medical image segmentation. *arXiv preprint arXiv:2102.04306* (2021)
3. Chen, R., Ma, Y., Chen, N., Lee, D., Wang, W.: Cephalometric landmark detection by attentive feature pyramid fusion and regression-voting. In: Shen, D., et al. (eds.) *MICCAI 2019. LNCS*, vol. 11766, pp. 873–881. Springer, Cham (2019). https://doi.org/10.1007/978-3-030-32248-9_97
4. Chen, R., et al.: Structure-aware long short-term memory network for 3d cephalometric landmark detection. *IEEE Trans. Med. Imaging* (2022)
5. He, K., Gkioxari, G., Dollár, P., Girshick, R.: Mask R-CNN. In: *Proceedings of the IEEE International Conference on Computer Vision*, pp. 2961–2969 (2017)
6. Hochreiter, S., Schmidhuber, J.: Long short-term memory. *Neural Comput.* **9**(8), 1735–1780 (1997)
7. Hurst, C.A., Eppley, B.L., Havlik, R.J., Sadove, A.M.: Surgical cephalometrics: applications and developments. *Plast. Reconstr. Surg.* **120**(6), 92e–104e (2007)
8. Kipf, T.N., Welling, M.: Semi-supervised classification with graph convolutional networks. *arXiv preprint arXiv:1609.02907* (2016)

9. Lang, Y., et al.: Automatic localization of landmarks in craniomaxillofacial CBCT images using a local attention-based graph convolution network. In: Martel, A.L., et al. (eds.) MICCAI 2020. LNCS, vol. 12264, pp. 817–826. Springer, Cham (2020). https://doi.org/10.1007/978-3-030-59719-1_79
10. Li, W., et al.: Structured landmark detection via topology-adapting deep graph learning. In: Vedaldi, A., Bischof, H., Brox, T., Frahm, J.-M. (eds.) ECCV 2020. LNCS, vol. 12354, pp. 266–283. Springer, Cham (2020). https://doi.org/10.1007/978-3-030-58545-7_16
11. Liu, Z., et al.: Swin transformer: hierarchical vision transformer using shifted windows. In: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 10012–10022 (2021)
12. Oh, K., Oh, I.S., Lee, D.W., et al.: Deep anatomical context feature learning for cephalometric landmark detection. *IEEE J. Biomed. Health Inform.* **25**(3), 806–817 (2020)
13. Payer, C., Štern, D., Bischof, H., Urschler, M.: Integrating spatial configuration into heatmap regression based CNNs for landmark localization. *Med. Image Anal.* **54**, 207–219 (2019)
14. Qian, J., Cheng, M., Tao, Y., Lin, J., Lin, H.: CephaNet: an improved faster R-CNN for cephalometric landmark detection. In: 2019 IEEE 16th International Symposium on Biomedical Imaging (ISBI 2019), pp. 868–871. IEEE (2019)
15. Raudaschl, P.F., et al.: Evaluation of segmentation methods on head and neck CT: auto-segmentation challenge 2015. *Med. Phys.* **44**(5), 2020–2036 (2017)
16. Ronneberger, O., Fischer, P., Brox, T.: U-net: convolutional networks for biomedical image segmentation. In: Navab, N., Hornegger, J., Wells, W.M., Frangi, A.F. (eds.) MICCAI 2015. LNCS, vol. 9351, pp. 234–241. Springer, Cham (2015). https://doi.org/10.1007/978-3-319-24574-4_28
17. Vaswani, A., et al.: Attention is all you need. *Advances in Neural Inf. Process. Syst.* **30** (2017)
18. Wang, C.W., et al.: Evaluation and comparison of anatomical landmark detection methods for cephalometric x-ray images: a grand challenge. *IEEE Trans. Med. Imaging* **34**(9), 1890–1900 (2015)
19. Wang, C.W., et al.: A benchmark for comparison of dental radiography analysis algorithms. *Med. Image Anal.* **31**, 63–76 (2016)
20. Wang, W., Chen, C., Ding, M., Yu, H., Zha, S., Li, J.: TransBTS: multimodal brain tumor segmentation using transformer. In: de Bruijne, M., et al. (eds.) MICCAI 2021. LNCS, vol. 12901, pp. 109–119. Springer, Cham (2021). https://doi.org/10.1007/978-3-030-87193-2_11
21. Xu, W., Xu, Y., Chang, T., Tu, Z.: Co-scale conv-attentional image transformers. In: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 9981–9990 (2021)
22. Yao, Q., Quan, Q., Xiao, L., Kevin Zhou, S.: One-shot medical landmark detection. In: de Bruijne, M., et al. (eds.) MICCAI 2021. LNCS, vol. 12902, pp. 177–188. Springer, Cham (2021). https://doi.org/10.1007/978-3-030-87196-3_17
23. Yueyuan, A., Hong, W.: Swin transformer combined with convolutional encoder for cephalometric landmarks detection. In: 2021 18th International Computer Conference on Wavelet Active Media Technology and Information Processing (ICCWAMTIP), pp. 184–187. IEEE (2021)
24. Yun, H.S., Hyun, C.M., Baek, S.H., Lee, S.H., Seo, J.K.: Automated 3D cephalometric landmark identification using computerized tomography. *arXiv preprint arXiv:2101.05205* (2020)

25. Zeng, M., Yan, Z., Liu, S., Zhou, Y., Qiu, L.: Cascaded convolutional networks for automatic cephalometric landmark detection. *Med. Image Anal.* **68**, 101904 (2021)
26. Zhong, Z., Li, J., Zhang, Z., Jiao, Z., Gao, X.: An attention-guided deep regression model for landmark detection in cephalograms. In: Shen, D., et al. (eds.) *MICCAI 2019*. LNCS, vol. 11769, pp. 540–548. Springer, Cham (2019). https://doi.org/10.1007/978-3-030-32226-7_60
27. Zhou, G.Q., et al.: Learn fine-grained adaptive loss for multiple anatomical landmark detection in medical images. *IEEE J. Biomed. Health Inform.* **25**(10), 3854–3864 (2021)
28. Zhou, H.Y., Guo, J., Zhang, Y., Yu, L., Wang, L., Yu, Y.: nnFormer: interleaved transformer for volumetric segmentation. *arXiv preprint* [arXiv:2109.03201](https://arxiv.org/abs/2109.03201) (2021)
29. Zhu, H., Yao, Q., Xiao, L., Zhou, S.K.: You only learn once: universal anatomical landmark detection. In: de Bruijne, M., et al. (eds.) *MICCAI 2021*. LNCS, vol. 12905, pp. 85–95. Springer, Cham (2021). https://doi.org/10.1007/978-3-030-87240-3_9